Homework 4
Elementary Linear Algebra
(MATH 2250-03), Spring, 2018

Check: 03/01/2019   Due: 03/04/2019

Section 2.2 - The Inverse of a Matrix

Let $A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$, let $B = \begin{bmatrix} 3 & -4 \\ 7 & -8 \end{bmatrix}$, and let $C = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$.

1. Compute $\left(AB\right)^{-1}$.

Recall that if A and B are invertible matrices, then $\left(AB\right)^{-1} = B^{-1}A^{-1}$. Using the fact that a 2×2 matrix M of the form $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $ad - bc \neq 0$ is invertible and $M^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, we compute $A^{-1} = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix}$ and $B^{-1} = \frac{1}{4}\begin{bmatrix} -8 & 4 \\ -7 & 3 \end{bmatrix}$. Therefore, our final answer is

$$\left(AB\right)^{-1} = B^{-1}A^{-1} = \left(\frac{1}{4}\begin{bmatrix} -8 & 4 \\ -7 & 3 \end{bmatrix}\right)\left(-\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix}\right) = \begin{bmatrix} 15/2 & -7/2 \\ 49/8 & -23/8 \end{bmatrix}.$$

2. Compute $\left(C^T\right)^{-1}$.

First, we compute C^{-1} by row reducing the augmented matrix $\left[\begin{array}{c|c} C & I \end{array}\right]$ to row echelon form. This produces

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & 7/2 & 3/2 & 1/2 \end{array}\right] \text{(Row reduce)}$$

The inverse of C is the right half of this augmented matrix, so, recalling that if C is an invertible matrix, then $\left(C^T\right)^{-1} = \left(C^{-1}\right)^T$, we may conclude by computing

$$\left(C^T\right)^{-1} = \left(C^{-1}\right)^T = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}^T = \begin{bmatrix} 8 & 10 & 7/2 \\ 3 & 4 & 3/2 \\ 1 & 1 & 1/2 \end{bmatrix}.$$

SECTION 2.3 - CHARACTERIZATIONS OF INVERTIBLE MATRICES

3. Let A and B be $n \times n$ matrices and suppose $BA = I$. Show that A is invertible and that $B = A^{-1}$.

Let $x \in \mathbb{R}^n$ be any vector such that $Ax = 0$. Then

$$x = Ix = \left(BA\right)x = B\left(Ax\right) = B0 = 0.$$

So, if $x \in \mathbb{R}^n$ is such that $Ax = 0$, then it must be the case that $x = 0$. That is, the homogeneous system $Ax = 0$ has only the trivial solution. So, A is row equivalent to the identity matrix. As an elementary row operation is equivalent to mulitiplication by the corresponding elementary matrix, and as elementary matrices are invertible, this yields that there is an invertible matrix $E = E_1 \ldots E_k$, with E_i an elementary matrix, such that $EA = I$. Hence, $A = E^{-1}$, so A is invertible and and $A^{-1} = E$. So, $B = BAA^{-1} = A^{-1}$ and we are done.

4. Let A and B be $n \times n$ matrices and suppose AB is invertible. Show that A is invertible and give a specific formula for the inverse of A in terms of $(AB)^{-1}$ and B.
(HINT: Use what you showed in problem 3.)

Using the associative property of matrix multiplication and the invertiblity of AB, we have $\left(AB\right)\left(AB\right)^{-1} = A\left(B\left(AB\right)^{-1}\right) = I$. Hence, the matrix $C = B\left(AB\right)^{-1}$ satisfies $AC = I$. Using the result of problem 4, it then follows that C is invertible and $A = C^{-1}$. So, noting that $AC = CC^{-1} = I$ and $CA = CC^{-1} = I$, it is clear that A is invertible and $A^{-1} = \left(C^{-1}\right)^{-1} = B\left(AB\right)^{-1}$.

SECTION 2.4 - PARTITIONED MATRICES

5. Suppose A, B, and C are $n \times n$ matrices, and let X, Y, and Z be $n \times n$ matrices such that

$$\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix}\begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

Find formulas for X, Y, and Z in terms of A, B, and C.
(HINT: Use what you showed in problem 3.)

Computing the product of the block matrices, we have

$$\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix}\begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} (XA+0B) & (X0+0C) \\ (YA+ZB) & (Y0+ZC) \end{bmatrix} = \begin{bmatrix} XA & 0 \\ (YA+ZB) & ZC \end{bmatrix}.$$

So, $\begin{bmatrix} XA & 0 \\ (YA+ZB) & ZC \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$. That is, X, Y, and Z must be such that $XA = I$, $YA + ZB = 0$, and $ZC = I$. Using the result of problem 3, it follows that A and C are invertible and that $X = A^{-1}$ and $Z = C^{-1}$. So, using this and the invertibility of A,

$$Y + C^{-1}BA^{-1} = \left(YA + C^{-1}B\right)A^{-1} = \left(YA + ZB\right)A^{-1} = 0A^{-1} = 0.$$

Therefore, we may conclude that $X = A^{-1}$, $Y = -C^{-1}BA^{-1}$, and $Z = C^{-1}$.

6. Let $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$ and let $b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$. Solve the equation $Ax = b$ by using the following LU factorization of A:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

First, we solve

$$Ly = b$$
$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$
$$\begin{bmatrix} y_1 \\ -y_1 + y_2 \\ 2y_1 + y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}.$$

Back substituting, this produces

$$y_1 = 2, \quad y_2 = -4 + y_1 = -2, \quad y_3 = 6 - 2y_1 = 2.$$

We then solve

$$Ux = y$$
$$\begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$$
$$\begin{bmatrix} 4x_1 + 3x_2 - 5x_3 \\ -2x_2 + 2x_3 \\ 2x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}.$$

Back substituting, this produces

$$x_3 = 1, \quad x_2 = (-1/2)(-2 - 2x_3) = 2, \quad x_1 = (1/4)(2 - 3x_2 + 5x_3) = 1/4.$$

So, we have found a solution to $Ux = y$ as desired, and as $Ax = LUx = Ly = b$, we may conclude that $x = \begin{bmatrix} 1/4 \\ 2 \\ 1 \end{bmatrix}$ is a solution to $Ax = b$